

GAWTAM CHITHRA RAMESH

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Summary

I am a robotics researcher spanning manipulation, motion planning, and perception, with hands-on ownership of projects from algorithm design through validation on physical hardware in industrial R&D settings (ABB, Ericsson). I build and deploy real-time systems bridging simulation and reality, and publish results in peer-reviewed venues.

Education

KTH Royal Institute of Technology

Master of Science in Systems, Controls, and Robotics (GPA 4.62/5.0)

Aug 2024 – Jun 2026

Stockholm, Sweden

Indian Institute of Technology Madras

Dual Degree (B.Tech Engineering Design and M.Tech Robotics) (GPA 7.9/10)

Aug 2019 – Jul 2024

Chennai, India

- Awarded (among 30 of 800+) the IIT Madras **Young Research Fellowship** based on remarkable research potential.
- Secured **97.64** Percentile in JEE Advanced and **99.10** Percentile in JEE Mains among **1.5M** candidates.
- Secured an All India Rank of **52** out of 50k in Undergraduate Common Entrance Examination for Design.

Skills : Python, C++, ROS/ROS2, MoveIt, Gazebo, MuJoCo, PyTorch, JAX, OpenCV, Docker, URDF/SDF, CUDA

Professional Experience

Master's Thesis: Generative Motion Planning for Manipulation

Jan 2026 – Jun 2026

ABB Robotics

Västerås, Sweden

- Designed a generative flow-matching motion planner guided at inference by Signal Temporal Logic robustness gradients.
- Steered the pretrained planner toward new task specifications at inference without retraining, cutting iteration time.
- Collaborating with ABB R&D to transfer the approach from benchmark tasks to long-horizon 6-DoF manipulation.

Device Research Intern: 3D Scene Graphs

Jul 2025 – Dec 2025

Ericsson Research

Lund, Sweden

- Integrated instance segmentation and tracking into Hydra, a real-time 3D scene-graph construction framework.
- Built a synchronized RGB-D inpainting pipeline that removes privacy-sensitive objects from scene graphs online.
- Benchmarked state-of-the-art inpainting models to balance reconstruction fidelity against real-time latency.

Graduate Robotics Intern: Modular Collaborative Robots

Dec 2022 – Jul 2023

Systemantics India

Bangalore, India

- Owned the real-time system controller for a 6-DoF cobot portfolio end to end, deployed on Linux PREEMPT-RT kernel.
- Handled inter-process communication over SocketCAN and FDCAN buses using semaphores and signal interrupts.
- Integrated a ROS2 stack with MoveIt motion planning and Gazebo simulation, authoring URDF and SRDF descriptions.

Research Experience

Autonomous Ultrasound Scanning via 3D Surface Reconstruction

Jan 2024 – May 2024

Nirav Patel, INSPIRE Lab, IIT Madras

Chennai, India

- Performed 3D torso surface reconstruction from 20 predefined RGB poses via multi-view stereo on a 5-DoF robot arm.
- Segmented the abdominal ROI from the point cloud and estimated probe approach angles from surface tangent vectors.
- Closed the perception-to-planning loop with ROS/MoveIt motion planning validated on physical hardware in simulation.

Target-Tracking Motion Planning for UAVs

Sep 2021 – Apr 2022

Prof. Satadal Ghosh, IIT Madras

Chennai, India

- Built 2D target-tracking and obstacle avoidance for UAVs via Proportional Navigation and velocity obstacles.
- Achieved roughly 20% faster trajectories with lower compute cost than classical aerospace motion planning algorithms.
- Awarded the IIT Madras Young Research Fellowship (top 30 of 800+ applicants) for outstanding research contribution.

Technical Projects

EKF-SLAM and State Estimation | EL2320, KTH

Mar 2025 – Jun 2025

- Implemented 2D EKF-SLAM from scratch with known and unknown data association via Mahalanobis distance gating.
- Extended filter state and covariance dynamically per new landmark discovery; resolved a matrix-consistency bug.
- Showed position error within 3σ under known association; divergence on symmetric maps revealed EKF overconfidence.

Multi-Variable Dynamics Modeling and H_∞ Control | EL2520, KTH

Jan 2025 – Mar 2025

- Modeled a four-tank process from first principles; identified min- and non-min-phase regimes through RGA analysis.
- Designed a decentralized decoupling controller and a robust H_∞ loop-shaping controller for the hydraulic process.
- Validated the H_∞ controller: 4s rise time, 20% overshoot, and 2% steady-state error in the minimum-phase regime.

Robot IK, Motion Planning, and Mapping | DD2410, KTH

Sep 2024 – Oct 2024

- Solved analytic IK for a 3-DoF arm and Jacobian-based IK for a 7-DoF manipulator using DH parameterization.
- Implemented Hybrid A* and RRT for a Dubins car; generated collision-free paths around circular and line obstacles.
- Developed a Behavior Tree mission planner for autonomous navigation, grasping, and recovery in ROS/Gazebo.

Lead Author Publications

- Robust Object-layer Construction of 3DSG Using Instance Segmentation (IMPROVE26)(**Best Paper Award**)
- Synchronized RGB-D Inpainting for Privacy-Aware 3D Scene Graph Construction.(**ICPR26, ECCV26 Workshop**)
- Scene Understanding in Deformable Object Manipulation via Taxonomy-Guided VLMs.(**IROS25 Workshop**)